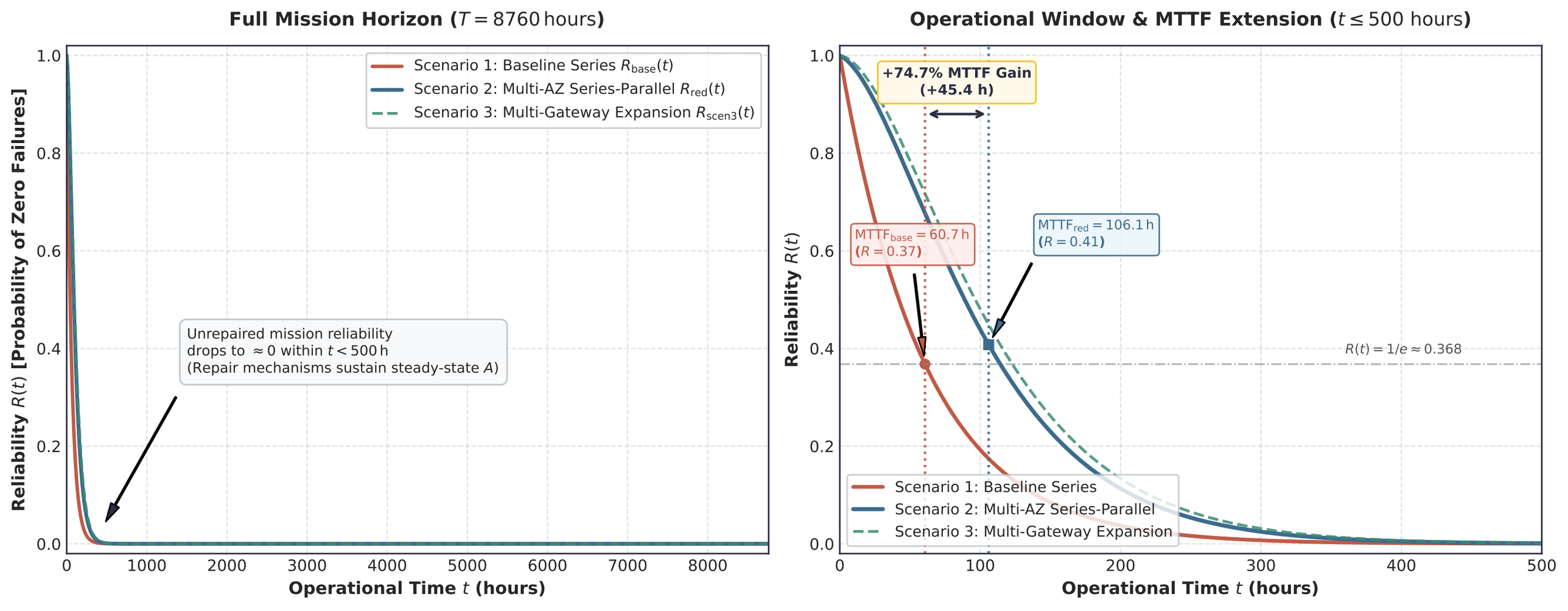
Full Mission Horizon ($T = 8760$ hours)
Reliability $R(t)$ [Probability of Zero Failures]
Operational Time $t$ (hours)
Scenario 1: Baseline Series $R_{base}(t)$
Scenario 2: Multi-AZ Series-Parallel $R_{red}(t)$
Scenario 3: Multi-Gateway Expansion $R_{scen3}(t)$
Unrepaired mission reliability
drops to $\approx 0$ within $t < 500$ h
(Repair mechanisms sustain steady-state $A$)
Operational Window & MTTF Extension ($t \leq 500$ hours)
Reliability $R(t)$
Operational Time $t$ (hours)
+74.7% MTTF Gain
(+45.4 h)
$\text{MTTF}_{base} = 60.7$ h
($R = 0.37$)
$\text{MTTF}_{red} = 106.1$ h
($R = 0.41$)
$R(t) = 1/e \approx 0.368$
Scenario 1: Baseline Series
Scenario 2: Multi-AZ Series-Parallel
Scenario 3: Multi-Gateway Expansion